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Adaptive Emotion Detection Using the Valence-Arousal-Dominance Model and EEG Brain Rhythmic Activity Changes in Relevant Brain Lobes

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ABSTRACT Contrary to popular belief that only the frontal lobe is concerned with emotions, recent neuroscience experiments show that in each emotional state, some of the brain lobes fired clearly whereas some did not. Unlike most of the previous works, which focused on choosing a fixed set of electrodes to detect emotions, this work presents a novel approach that consists of investigating the cerebral activity while experiencing emotions to identify the brain lobes that are showing significant and relevant changes. The identification process of relevant brain lobes is adaptive as the brain activity varies from one person to another and differs from one lobe to another during the same emotional state. The electrodes of the selected brain lobes will be tuned before being used as resources to extract the features that are required for the proposed three phases classification process. Using a clustering technique, the tuning process excludes every electrode that hardly separates between relevant and irrelevant rhythmic brain changes. The results show that the proposed method outperforms all previous approaches. Compared to the best performance obtained by previous studies, the proposed method enhanced the accuracy of both valence and arousal dimensions by 5%. Besides, the accuracy of the dominance dimension was improved by 2%.

INDEX TERMS Brain–computer interfaces, EEG, machine learning, multi-class classification, one-class classifier, VAD model.

I. INTRODUCTION

Emotions are human responses to an object or an event in the environment [1]. They affect different aspects of human life, such as attention, memorization, achieving goals, awareness of priority, knowledge motivation, communication with others, learning development, mood status, and effort motivation [2], [3]. Over the past decade, two main approaches have been developed to analyze and classify emotions: the locationist approach and the construction approach. The first approach assumes that there is a specific brain structure and pattern for each emotion. Conversely, the second approach allows the study of emotions based on broader aspects. It defines several dimensions to create an effective framework to study and classify emotions.

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Prior studies adopting the constructionist approach deal with emotions using an affective valence model or a motivational direction model [3], [4]. Affective valence implies categorizing emotions as positive or negative. Positive emotions include happiness, joy, excitement, and interest, whereas negative emotions include sadness, anger, disappointment, and anxiety. Conversely, the motivational direction category has different characteristics. In this approach, emotions are classified as appetitive (called approach related) or aversive (called withdrawal related) to the stimulus. Appetitive emotions cause the person to consider a situation (either pleasant or unpleasant) with positive or negative feelings, for example, interest, anger, or joy. Aversive emotions cause the person to withdraw from the situation.

A novel multidimensional constructionist model links emotions with a wider dynamic neural network of brain activities. Lindquist in 2013 [5], Barrett and Lindquist in 2012

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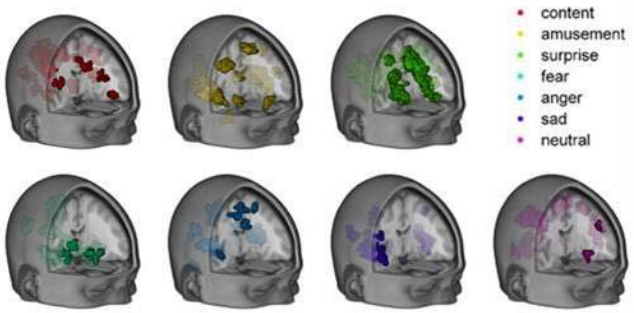


FIGURE 1. Functional magnetic resonance imaging scans of the brain activity during different emotional sates [11].

[6], Barrett and Satpute in 2013 [7] studied emotions with general domain activities instead of focusing on a specific brain center for each emotion. This model depends on several dimensions to create an effective framework for studying emotions instead of studying groups of discrete emotions. There is no limitation to the number of dimensions. Nevertheless, there are three dimensions used in most studies. This is called the valence-arousal-dominance (VAD) constructionist model. The VAD dimensions are not reflected in specific regions of the brain. They produce different neuron patterns varying from one level to another [3].

Inevitably, electroencephalography (EEG) is a useful technique to explore neural network activities for any emotional state [2] as EEG measures show the electrical activity during an emotional experience. In general, for the valence dimension, its intensity is detected by studying alpha and beta bands; positive emotions, such as happiness, show an increase in the alpha signal in the frontal lobe and an increase in the beta signal in the right parietal lobe of the brain, unlike the opposite emotions [8]. In contract, arousal can be noticed with a decrease in the alpha signal and an increase in the beta signal in the parietal lobe of the brain. The dominance dimension can be measured by noticing an increase in the activity ratio between alpha and beta signals as well as an increase in the beta power in the parietal lobe [8]. Neuroscience experiments on the human brain show that all regions (or almost all) exhibit a ''firing'' activity during emotional experiences [9], [10]. Moreover, contrary to popular belief that only the frontal lobe is concerned with emotions, a recent study in neuroscience published in 2016 [11] showed that for each emotion, some of the brain lobes fired clearly whereas some did not. The activity is unique in each emotion, as illustrated in the scans below. Fig.1 shows the functional magnetic resonance imaging (FMRI) scans of the brain activity during different emotional sates.

The emotion detection process, in almost all previous works, was usually the same for all subjects, whereas neuroscience studies [3], [12], [13] mentioned that the activity is sophisticated and unique in the brain of each person. Moreover, the brain activity is different in the left and right lobes during the same emotional experience [3], [9]. Thus, there is a requirement to study emotions subjectively as a prerequisite step to an objective conclusion. Studying emotions subjectively results in an objective understanding in this field. In this work, we applied an adaptive processing of the brain activity to classify emotions. The approach enables the building of a processing model that is specific to every subject. In this work, all essences of the brain activities were considered to choose signal resources and to extract relevant features before the classification step.

II. OVERVIEW OF THE VALENCE-AROUSAL- DOMINANCE CONSTRUCTIONIST MODEL

In the VAD model, emotions are studied based on broader aspects instead of a specific brain structure and pattern for each emotion. This model links emotion with the wider dynamic neural network of brain activities [3], [10], [14], [15]. The VAD model defines three dimensions: valence, arousal, and dominance. Every dimension has three levels. In the valence dimension, the three levels are neutral, negative, and positive. In the arousal dimension, they are neutral, passive, and active. In the dominance dimension, the levels are neutral, low dominance, and high dominance. The combination of the three labels (levels) configures the emotion label. Consequently, 27 emotions could be classified depending on the dimension labels.

For example, as shown in Fig.2a, the triplet (negative, passive, high) corresponds to the depressed emotional state. In Fig.2b, the triplet (positive, active, low) in the VAD model corresponds to the excited emotional state.

III. RELATED WORK

As mentioned earlier, all previous studies depended on a fixed number of electrodes as a feature extraction resource [16]–[19]. Regardless of the variety of the features chosen from one study to another, the process of emotion detection and classification started with signal preprocessing, followed by feature extraction, and finally the application of a feature vector to a classifier. In this section, we will discuss the previous valuable work in the emotion detection field using EEG signals.

A series of investigations have studied emotions based on EEG signals [8], [20], [21]. Many of them used two emotional classes (this differs from one study to another). In one such study, two emotional classes were investigated: calm and negatively excited (VAD approach) [22]. Features such as fractional dimensionality, correlation dimension, the Lyapunov exponent, and certain statistical features on wavelet coefficients were used. The statistical features were mean, average power, and wavelet entropy. The defined features were extracted using a group of chosen EEG electrodes: FP1, FP2, T3, T4 and Pz [22]. In addition to the quantitative analysis, a qualitative analysis was applied to select features using the t test. The classifiers used were linear discriminating analysis (LDA) and support vector machine (SVM). The accuracy was 81% and 84% for LDA and SVM, respectively. In this work, the valence and arousal values of the two statuses

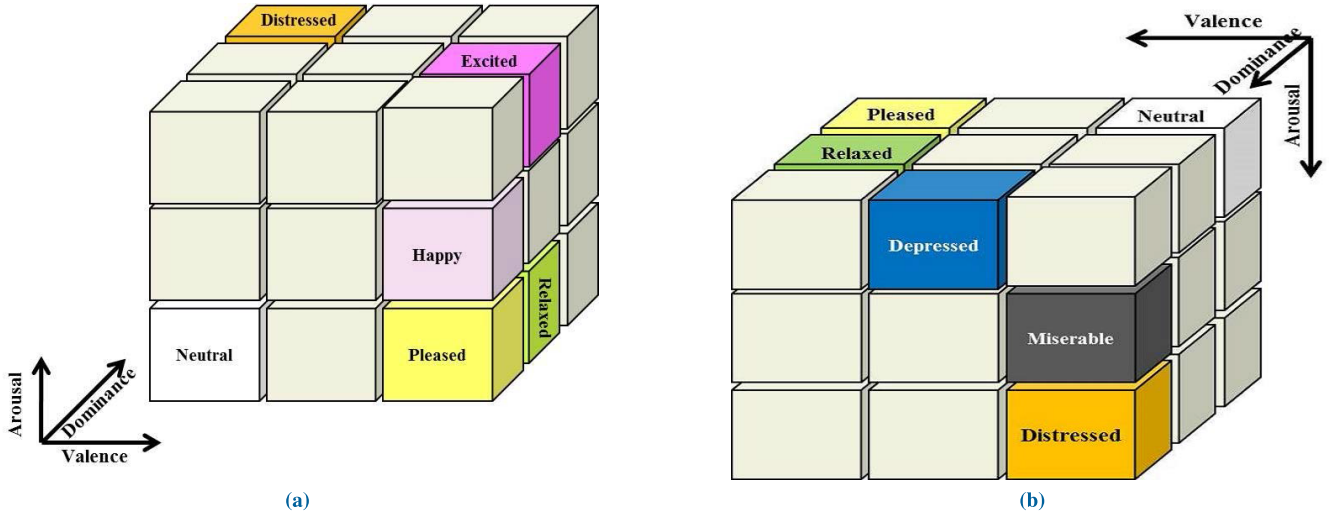


FIGURE 2. (a), (b) 3D emotions cube (viewed from different angles).

(calm and negatively excited) were not crossed. The values were as follows: arousal < four and valence was between four and six in the calm state, and arousal > five and valence < three in the negative state. Classifying these categories will be more difficult with additional emotions.

Another study classified valence and arousal states with a two-class classification (low or high) [23]. EEG features such as correlation analysis of frequency and baseline powers for six subjects using music video stimuli were used. Then, both power spectral density (PSD) and common spatial patterns were applied for feature selection on 32 EEG channels. Finally, SVM was applied to classify these categories. The results were 55.7% for arousal and 58.8% for valence. In this work, the self-assessment emotion labeling technique was used. No numeric values were used to specify the levels of valence and arousal.

As in the previous study, in another work, arousal and valence were classified into two levels (high and low) using different combinations of features, such as time-domain, frequency-domain, multi-electrode features, and connectivity features (mutual information and Pearson correlation) [16]. The number of features was large; therefore, the authors applied maximum relevance minimum redundancy and principal component analysis (PCA) for hybrid dimension feature reduction. The accuracy of their work was 74.3% and 77.2% for arousal and valence, respectively, using the random forest classifier. Further, they recorded accuracy values of 72.1% (arousal) and 76.1% (valence) using SVM.

Further, another study [18] stated that emotion classification depends on the (low/high) values of valence and arousal. The process of their work calculated spectral power and mutual information of all electrodes using the Dataset for Emotion Analysis using Physiological Signals (DEAP) dataset. The mutual information was calculated from filtered and unfiltered EEG signals in the four bands: alpha, beta, theta, and gamma. Subsequently, the authors reduced the dimensionality by applying PCA. The average accuracy results for classifying high/low levels of arousal were 65.6%, 65%, and 67.7% for naïve Bayes, SVM (radial basis function kernel), and SVM (sigmoid kernel), respectively. Moreover, for valence, the average accuracies were 68%, 63.3%, and 69.2%, respectively, for the same classifiers.

A study in 2017 [24] tested different EEG features to classify valence and arousal into low and high levels. They calculated features from the empirical mode decomposition (EMD) domain. The EMD decomposes signals into different frequencies and band-limited features to result in a nonlinear sequence of data. From this domain, the authors chose sample entropy, discrete wavelet transform with differential entropy in the beta band, and the decomposition as features in the gamma band. They extracted features from eight electrodes (F7, Fp1, Fp2, F8, P9, P7, T8, and T7). When they applied the SVM classifier with the chosen features, the authors recorded average accuracy values of 69.10% for valence and 71.99% for arousal.

Improving the accuracy of a multiclass emotion detection system is always a problem to be resolved. One study discussed the multiclass classification technique as a method to improve the accuracy . First, the researchers extracted PSD from 12 symmetric pairs of electrodes. Then, using SVM, the researchers applied two approaches: one-vs-one and all-vs-all. For the first approach, binary SVM was used; moreover, a library for SVM was used for the latter approach. In addition, a model-based approach was used. The classification was divided into two levels: valence and arousal. The valence level (positive/negative) specifies if the emotions are anger/sadness or joy/pleasure. The arousal (active/passive) level specifies if the emotion is anger or sadness when the valence is negative and joy or pleasure if the valence is positive. The study obtained interesting results: one-vs-one scored the best average accuracy results (92.57%) for 26 subjects. Emotion detection becomes more reliable

when using the VAD model with three dimensions. As in [17], emotions are classified using different methods depending on the three dimensions. The authors chose spectral entropy as a feature from 32 EEG electrodes. They used the DEAP dataset to test SVM, K-nearest neighbor, multilayer perceptron, and meta multiclass with 10-fold cross validation. The average accuracies were 81.45%, 74.37%, 57.74%, and 75.94%, respectively, for classifying 13 emotions. The authors generated the emotion label by mapping it to clustering groups (related to the three dimensions) as they called it a multimodal fusion framework [17].

IV. TERMINOLOGY AND ANNOTATIONS

1) Let us define F as the set of frequency bands of brain waves. The frequency bands that we consider in this work are:
   - Theta (θ): 4 to 7 Hz
   - Alpha (α): 8 to 12 Hz
   - Beta (β) : 12 to 36 Hz
   - Gamma (γ): 36 to 42 Hz
2) We defined B as the set of brain lobes. The brain lobes, which we consider in this study, are the frontal, central, temporal, occipital, and parietal lobes. Every brain lobe is composed of two parts: the left and right brain lobes.
3) An electrode denoted as "e" is an electrical conductor used to acquire the brain signals. An electrode e has two main characteristics:
   - A **label**: It is the name of the electrode. It is a letter that indicates the cerebral region to which the electrode belongs.
   - A **serial number**: It is a unique two-digit integer number within the cerebral region. If this number is odd, it implies that the electrode is in the left brain lobe. Otherwise, it implies that the electrode is in the right brain lobe.
4) We defined E as the set of electrodes e of a cap.
5) Let us consider the function ε as follows:
   - $\varepsilon : E \rightarrow B$
     $\varepsilon(e) = b$ such that e belongs to the brain lobe b.
   - $\varepsilon^{-1} : B \rightarrow P(E)$
     $\varepsilon^{-1}(b) = \{e_i \in E$ such that e_i belongs to the brain lobe $b\} \subset E$. Given a brain lobe b, this function returns the set of electrodes that belongs to b.
6) An emotion, denoted as η, is the mental state resulting from a given internal or external event.
7) We defined Ψ as the set of emotional states. The emotional states that we considered in this work are "happy", "pleased", "relaxed", "excited", "neutral", "calm", "distressed", "miserable", and "depressed".
8) A neutral state, denoted as η_0, is the mental state that corresponds to the "Neutral" emotional state.
9) We defined ς as the set of human beings called subjects.
10) A trial, denoted as t, is a set of signals that were recorded using a set of electrodes E and that corresponded to the brain activity of a given subject s during a given emotional state η.
11) Let us define Ω as the set of trials.
12) We defined D as the set of dimensions of the VAD model, which was composed of the following dimensions:
    - The dimension **valence**: It mentions whether the feeling is pleasant or not.
    - The dimension **arousal**: It represents the degree of excitement.
    - The dimension **dominance**: It represents how much the excitement is controlled.
13) Let us define L as the set of dimension levels composed of the following levels:
    - **Neutral**, **positive**, and **negative**: They are levels of the dimension valence. Negative and positive levels are called the lower and upper levels of the dimension valence, respectively.
    - **Neutral**, **active**, and **passive**: They are levels of the dimension arousal. Passive and active levels are called the lower and upper levels of the dimension arousal, respectively.
    - **Neutral**, **high**, and **low**: They are levels of the dimension dominance. Low and high levels are called the lower and upper levels of the dimension dominance, respectively.

V. PROPOSED METHOD

A. PRELIMINARIES

We defined $\Phi(d)$ as the subset of the levels that are appropriate to a given dimension d. $\Phi(d)$ is defined using the following expression:

- Φ: $D \rightarrow P(L)$
  $\Phi(d) = \{l_i \in L$, such that l_i corresponds to a level of the dimension d $\} \subset L$.

We defined the map function as follows:

- map: $\Phi(valence) \times \Phi(arousal) \times \Phi(dominance) \rightarrow \Psi$. For a given level x of the valence dimension, level y of the dimension arousal, and level z of the dimension dominance, this function determines, according to the VAD model, the emotional state that corresponds to the triplet $< x, y, z >$.

We defined $g(l)$, $h(l)$, and $k(l)$ as selectors that determine the set of different emotional states that correspond to the level l in the arousal, dominance, and valence dimensions, respectively. They are correspondingly defined with the following expressions:

- g: $\Phi(valence) \rightarrow P(\Psi)$

$$g(l) = \begin{cases} \varnothing \ if \ l \notin \Phi(valence) \\ \bigcup_{<x,y> \in \Phi(arousal) \times \Phi(dominance)} map(l, x, y) \end{cases}$$

- h: $\Phi(arousal) \rightarrow P(\Psi)$

$$h(l) = \begin{cases} \varnothing \ if \ l \notin \Phi(arousal) \\ \bigcup_{<x,y> \in \Phi(valence) \times \Phi(dominance)} map(x, l, y) \end{cases}$$

- k: $\Phi(dominance) \rightarrow P(\Psi)$

$$k(l) = \begin{cases} \varnothing \ if \ l \notin \Phi(dominance) \\ \bigcup_{<x,y> \in \Phi(valence) \times \Phi(arousal)} map(x, y, l) \end{cases}$$

We defined the function $\mathcal{M}$, which determines for a given dimension d and a given level l, the set of emotional states that correspond to level l of d.

$$\mathcal{M}(d, l) = \begin{cases} g(l) \ if \ d = valence \\ h(l) \ if \ d = arousal \\ k(l) \ if \ d = dominance \end{cases}$$

We considered the function τ as follows:

- $\tau: \Omega \rightarrow \varsigma \times \Psi$
  $\tau(t_i) = (s, \eta)$
  The τ function associates every trial to a given subject and a given emotional state.
- $\tau^{-1} : \varsigma \times \Psi \rightarrow P(\Omega)$
  $\tau^{-1}(s, \eta) = \{t_i \in \Omega, \text{ such that } \tau(t_i) = (s, \eta)\} \subset \Omega$.
  It returns the subset of trials t_i recorder from the subject s during the emotional state η.

We defined the function σ as follows:

- $\sigma: \Psi \rightarrow P(\Omega)$
  $\sigma(\eta) = \bigcup_{si \in S} \tau^{-1}(s_i, \eta)$

Given an emotional state η, this function returns the subset of trials that have been recorded from different subjects during the emotional state η.

We defined the function δ as follows:

- $\delta: \varsigma \rightarrow P(\Omega)$
  $\delta(s) = \bigcup_{\eta_i \in \Psi} \tau^{-1}(s, \eta_i)$

Given a subject s, this function returns the subset of trials recorded from subject s during different emotional states.

B. MEASURING BRAIN RHYTHMIC ACTIVITIES

The variation of the rhythmic brain activity that is observed in a given electrode e during a trial t, in the frequency band f, which is denoted as $\Delta(e \in E, t \in \Omega, f \in F)$, is defined as the percentage of decrease or increase in power in relation to a reference power. This variation is measured according to the following expression (eq 1):

$$\Delta(e \in E, t \in \Omega, f \in F) = \frac{Power(e, t, f) - ref_Power}{ref_Power} \quad (1)$$

$Power(e, t, f)$ is the power energy of the electrode e in the frequency band f during the trial t. It is computed using a frequency-domain representation of the trial t by applying the fast Fourier transformation. It is measured according to the following expression (eq 2):

$$Power(e, t, f) = \sum_{k=lower(f)}^{upper(f)} kX(t, f)[e, k] \quad (2)$$

where

$$X(t, f)[j, k] = \sum_{n=0}^{N-1} x_n[j] e^{\frac{i2\pi}{n} kn} \ and \ N = \frac{\text{sampling rate}}{2}$$

C. SELECTION OF SENSITIVE BRAIN LOBES

Our proposed method is inspired from a study in neuroscience published in 2016 [11] showing FMRI scans of the brain activity during different emotional states. It appears that for each emotion, some of the brain lobes fired clearly whereas some did not. The activity is unique in each emotion, as the scans demonstrated. Based on this study, our objective is to determine the brain lobes that are most probably sensitive to emotions corresponding to each subject in every frequency band. Knowing that the folding of the cortex differs between any two persons, even monozygotic twins, we strongly believe that brain lobes that are sensitive to emotions differ from one person to another.

Definition 1: Let us consider a subject s, a dimension d, a level l of d, and a frequency band f. The brain lobe b of the subject s is sensitive to emotions of the level l of d with respect to the frequency band f, if the probability of getting significant brain activity changes (variations) in the brain lobe b of the subject s with respect to the frequency band f when emotions of level l of dimension d happen, which is denoted as $p(s, d, l, f, b)$, exceeds a given threshold called $\alpha_threshold$.

The probability $p(s, d, l, f, b)$ is calculated as follows (eq 3):

- $p : \varsigma \times D \times L \times F \times B \rightarrow [0, 1]$

$$p(s, d, l, f, b) = \frac{\| \omega(s, d, l, f, b) \|}{\| \varphi(s, d, l) \|} \quad (3)$$

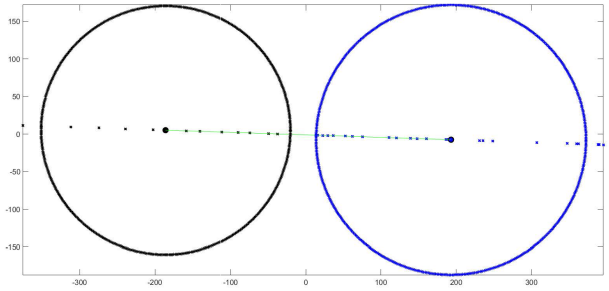
Here:

- $\| \varphi(s, d, l) \|$ and $\| \omega(s, d, l, f, b) \|$ are the cardinality of $\varphi(s, d, l)$ and $\omega(s, d, l, f, b)$, respectively.
- $\varphi(s, d, l)$ is the set of trials recorded from s during emotions that correspond, according to the VAD model, to the level l of the dimension d. $\varphi(s, d, l)$ is defined as follows (eq 4):

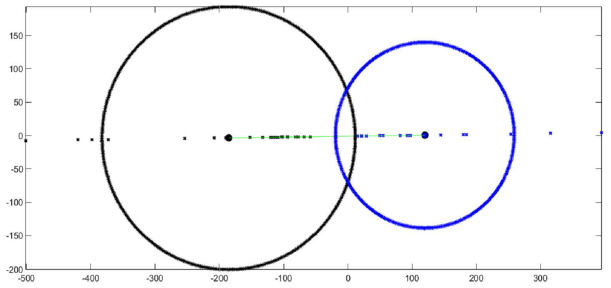
$$\varphi : \varsigma \times D \times L \rightarrow P(\Omega)$$
$$\varphi(s, d, l) = \delta(s) \cap \bigcup_{\eta_i \in \mathcal{M}(d,l)} \sigma(\eta_i) \quad (4)$$

- $\omega(s, d, l, f, b)$ is the subset of $\varphi(s, d, l)$, where variations in the frequency band f of the brain activity in the brain lobe b of the subject s are significant.

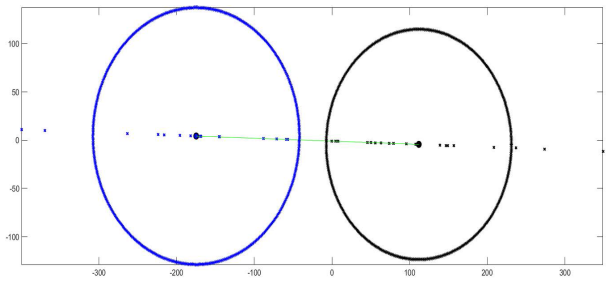
"Definition 1" aims to select in every frequency band f the brain lobes that have significant variations. Let's specify now the condition that variations in a given brain lobe with respect to a given frequency band f are to be considered significant.



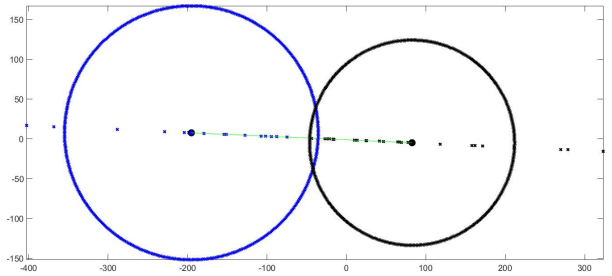
(a) Subject_1, alpha, Valence (positive), electrode: F4



(b) Subject_1, alpha, Valence (positive), electrode: AF4



(c) Subject_1, alpha, Valence (positive), electrode: O1



(d) Subject_1, alpha, Valence (positive), electrode: T8

FIGURE 3. **Example of feature tuning, clearly, both electrodes AF4 and T8 will be excluded from the feature sources in this step.**

Definition 2: Changes (or variations) in the frequency band f of the brain activity in the brain lobe b of a given subject s are significant if the average variation in f of the brain activity in all electrodes of b exceeds a given threshold called $\beta_threshold$.

Based on Definition 2, the set $\omega(s, d, l, f, b)$ is specified by the following expression (eq 5):

- $\omega : \varsigma \times D \times L \times F \times B \rightarrow P(\Omega)$

$$\omega(s, d, l, f, b) = \{t_i \in \varphi(s, d, l) \text{ such that } Avg_\Delta(b, t_i, \mathcal{M}_0^{s,d}, f) \geqslant \beta_threshold\} \quad (5)$$

Here

- $\mathcal{M}_0^{s,d}\varphi(s, d, Neutral)$ is the set of trials recorded from s during emotions that correspond to the neutral level of dimension d.
- $Avg_\Delta(b, t_i, \mathcal{M}_0^{s,d}, f)$ is the average variation in f of the brain activity in all electrodes of b. $Avg_\Delta(b, t_i, \mathcal{M}_0^{s,d}, f)$ is computed in relation to $\mathcal{M}_0^{s,d}$ according to the following expression (eq 6):

$$Avg_\Delta(b, t_i, \mathcal{M}_0^{s,d}, f) = \frac{\sum e_i \in \varepsilon^{-1}(b) |\Delta(e_i, t, Ref_power(e_i, \mathcal{M}_0^{s,d}, f), f)|}{\| \varepsilon^{-1}(b) \|} \quad (6)$$

Here, $Ref_power(e_i, \mathcal{M}_0^{s,d}, f)$ is the reference power for the electrode e_i. It is the average power in the frequency band f of the electrode e_i calculated using $\mathcal{M}_0^{s,d}$ according to the following expression (eq 7):

$$Ref_power(e_i, \mathcal{M}_0^{s,d}, f) = \frac{\sum_{t_j \in \mathcal{M}_0^{s,d}} Power(e_i, t_j, f)}{\| \mathcal{M}_0^{s,d} \|} = \frac{\sum_{t_j \in \varphi(s,d,neutral)} Power(e_i, t_j, f)}{\| \varphi(s, d, neutral) \|} \quad (7)$$

Therefore, we can determine for every subject and in every frequency band the set of brain lobes that are sensitive to emotions that correspond to a given level of a given VAD dimension using the function ϱ, which is defined as follows:

- $\varrho : \varsigma \times D \times L \times F \rightarrow P(B)$

$$\varrho(s, d, l, f) = \begin{cases} \varnothing \text{ if } l \notin \Phi(d); \\ \{b_i \in B \text{ such that } p(s, d, l, f, b_i) \geqslant \alpha_threshold\} \subset B \end{cases}$$

D. SELECTION OF RELEVANT ELECTRODES

Once the brain lobes that are sensitive to emotions are determined, we strongly believe that their corresponding electrodes must be investigated and analyzed deeply to detect the electrodes that are relevant to the emotions.

Definition 3: Let us consider a subject s, dimension d, level l of d, and frequency band f. The electrode e is relevant to emotions of the level l of d with respect to the frequency band f, if the probability of obtaining significant brain activity changes in e with respect to f when emotions of level l of dimension d occur, which is denoted as $p'(s, d, l, f, e)$, exceeds a given threshold called $\alpha'_threshold$. The probability $p'(s, d, l, f, e)$ is calculated as follows (eq 8):

$$\rho' : \varsigma \times D \times L \times F \times E \rightarrow [0, 1]$$
$$\rho'(s, d, l, f, e) = \frac{\| \gamma(s, d, l, f, e) \|}{\| \varphi(s, d, l) \|} \quad (8)$$

TABLE 1. The final decision criteria for each dimension classifier.

Final Decision Criteria		Classifier of Upper level	
		(target) label	(outlier) label
Classifier of Lower level	(target) label	*Error*	Lower level of the dimension
	(outlier) label	Upper level of the dimension	Neutral level of the dimension

Here

- $\| \varphi(s,d,l) \|$ and $\| \gamma(s,d,l,f,e) \|$ are the cardinality of $\varphi(s,d,l)$ and $\gamma(s,d,l,f,e)$, respectively.
- $\varphi(s,d,l)$ is the set of trials recorded from s during emotions that correspond, according to the VAD model, to the level l of dimension d.
- $\gamma(s,d,l,f,e)$ is the subset of $\varphi(s,d,l)$, where variations in the frequency band f of the brain activity in the electrode e of the subject s are significant, i.e., it exceeds the average variation in f of the brain activity, denoted $Avg_var(s,d,l,f,e)$, in the brain lobe b which the electrode e belongs to.

  The $\gamma(s,d,l,f,e)$ is defined as follows (eq 9):
- $\gamma : \varsigma \times D \times L \times F \times E \rightarrow P(\Omega)$

$$\gamma(s,d,l,f,e) = \{t_i \in \varphi(s,d,l) \text{ such that } |\Delta(e,t_i,Ref_power(e,\mathcal{M}_0^{s,d},f),f)| \geqslant Avg_var(s,d,l,f,e)\} \quad (9)$$

where

$$Avg_var(s,d,l,f,e) = \frac{\sum_{t_i \in \varphi(s,d,l)} Avg_\Delta(\varepsilon(e),t_i,\mathcal{M}_0^{s,d},f)}{\| \varphi(s,d,l) \|} \quad (10)$$

Thus, we can define an electrode selector, denoted as Γ, that determines for a given subject, in a given frequency band, the set of electrodes that are relevant to emotions that correspond to a given level of a given VAD dimension. The selector Γ is defined as follows (eq 11):

$$\Gamma : \varsigma \times D \times L \times F \rightarrow P(E)$$
$$\Gamma(s,d,l,f) = \{e_i \in \bigcup_{b_j \in \rho(s,d,l,f)} \varepsilon^{-1}(b_j), \textit{such that } \times \rho'(s,d,l,f,e_i) \geqslant \alpha'_threshold\} \quad (11)$$

TABLE 2. Mapping dimension labels to an emotion label.

Label			Distinct emotion
Valence	Arousal	Dominance	
1	2	1	happy
1	2	2	pleased
1	3	2	relaxed
1	1	1	excited
2	2	2	neutral
2	3	2	calm
3	1	2	distressed
3	2	3	miserable
3	3	3	depressed

TABLE 3. Relevant electrodes of the valence dimension.

Level	Subject	Frequency Bands			
		Alpha	Beta	Theta	Gamma
Positive	S_3	**O2**	O1, O2, P8	O2	**O1**
	S_5	T8	**T7**, O2	**Fp1, F3, FC5**, FC1, Fp2, **F8, FC2**, P8, PO4	T8, O1
	S_8	**AF3, F3,** FC1, C4, CP2, T7, T8, **O1**, O2, **P3**	Fp2, AF4, FC6, CP5, **CP1**, O1, O2, P8, PO4	Fp1, **AF3**, F3, FC5, **FC1**, T7, T8, O1, O2, **P4, P8, PO4**	C3, CP5, **CP1**, C4, CP6, **CP2**, T7, O2, P3, P7, PO3, P4, **P8**
	S_23	O2	**T7**	**Fp1**, CP6, O2	**Fp1, AF3**, F3, **F7, FC5, FC1, F8, FC6,** FC2, C3, CP1, C4, CP6, **CP2, T8**, O1, O2, **PO3**, P4,P8, PO4
	S_30		Fp2, **AF4,** FC2, T8, P3, PO3, P4, P8, PO4		**Fp2**, FC6, FC2, CP2, P4, PO4
Negative	S_3	PO3	AF3, F3, FC1, CP5, T7, O1, P8, PO4	O2	F3, **O1**
	S_5	T8	P3, P7	**Fp1, F3, F7, FC5,** FC1, **F8, FC2**, P8, PO4	FC5, **F8**
	S_8	**F3, F7,** T7, T8	T7, O1	Fp1, **AF3**, F3, F7, FC5, **FC1, Fp2, F4,** F8, FC6, **FC2, C3,** CP5, **CP1**, T7, O1, **P4, P8, PO4**	**Fp2,** F4, F8, FC6, **FC2**, C3, **CP1**, C4, CP6, **CP2**, T8, O1
	S_23	CP5, CP6, O2, P4	O2	O2	Fp2, F4, **F8, FC6**, FC2, **CP5**, CP1, T7
	S_30	T8	T8		CP6, **T7, O1**, O2

E. FEATURES SELECTION AND TUNING

For every level l of a given dimension d, in every frequency band f, we identified a set of electrodes that were relevant to emotions that correspond to the level l of d. The changes in the rhythmic brain activity in these electrodes in the alpha, beta, theta, and gamma frequency bands were used as features to classify emotions. These features were analyzed and tuned to exclude those that correspond to significant but not distinct changes in rhythmic brain activity when emotions happen. This tuning process excluded electrodes that have similar firing (in terms of value and direction) in both upper and lower levels of the corresponding dimension. Let us consider a subject s, a dimension d, and a frequency band f. Let us also consider an electrode e that belongs to $\Gamma(s,d,Upper_level(d),f) \cap \Gamma(s,d,Lower_level(d),f)$. We clustered the changes of the brain activity recorded in the electrode e into two clusters: Upper- and lower-level values. The Upper-level values correspond to the changes of the brain activity in the electrode e during emotions at the upper level of the dimension d. The lower-level values correspond to

TABLE 4. Relevant electrodes of the arousal dimension.

Level	Subject	Frequency Bands			
		Alpha	Beta	Theta	Gamma
Active	S_3	Fp1, AF3, F3, F7, AF4, C4, **CP6, CP2, O2, P4,** P8	FC1, O1, O2	AF4, F8, **FC2**, CP6, CP2, T7, T8, O1, O2, **P7,** PO3, P4, **P8**	**Fp1, AF3**, F3, F7, FC1, **AF4**, **O1**, O2, PO3, **P4**, P8
	S_5	O1	**T7**, O2	T8	
	S_8	**AF3, F3,** FC1, C4, CP2, T7, T8, **O1**, P4, **PO4**	Fp2	Fp1, **AF3**, F3, F7, **FC1**, **Fp2**, AF4, **F4**, F8, FC6, **FC2, C3, CP1**, C4, **CP2**, T7, T8, O1, O2, **P4, PO4**	Fp1, AF3, F3, **F7, FC5,** FC1, Fp2, F4, FC6, **FC2**, C3, CP5, **CP1**, C4, CP6, **CP2**, T7, T8, O1, O2, P3, P7
	S_23	Fp1, AF3, **F3**, F7, FC5, **FC1**, C3, CP5, CP1, T8, O1, P4, P8, **PO4**	**AF4**, F4, FC6, FC2, T8, O2, P8, PO4	C4, CP6, CP2, T7, O1, O2, P4, P8	**Fp1, AF3**, F3, **F7, FC5**, O1, O2, P3, P7, **PO3**, P4, P8, PO4
	S_30	O1, **P3, P7**, PO3	Fp1, **CP6**, T8	Fp1, FC5, C3, T7, O2	AF4, FC6, FC2, T8, O2
Passive	S_3	F7, FC1, AF4, **O2**	T7, O2	O2	**O1**, O2
	S_5	FC5, O1, P8	**C3,** CP5, CP1, **T7, T8,** O2, **P4**, P8, PO4		**P4**
	S_8	Fp1, **AF3, F3, F7,** FC1, T7, **O1,** P4, **PO4**	Fp2, T7	F4, **FC2**, C4, T7, T8	C3, T8, O1
	S_23	CP1, T8	**C3**, CP1, **T7**, P8, PO4	**Fp1**, F7, FC5, FC1, **Fp2**, AF4, F8, FC6, C4, CP6, O2	**Fp1, F7, FC5, T8,** O1, O2, P4, P8
	S_30		Fp1, C3, T8	Fp1, FC5, T7, T8, O2	AF4, **F8**, P3, PO3

TABLE 5. Relevant electrodes of the dominance dimension.

Level	Subject	Frequency Bands			
		Alpha	Beta	Theta	Gamma
Low	S_3	O1			
	S_5	O2			O1
	S_8	**AF3, F3**, FC1, C4, T8, O2	O2	**F4,** FC6, **FC2**, T7	**Fp2**, AF4, F4, F8, C3, CP5, **CP1**, C4, CP6, **CP2**, T7, T8, O2, P4, **P8**, PO4
	S_23	CP5, CP1, O1, O2, P7	**T7**		CP1
	S_30	**P3**, PO3	**O1**	**AF4**, F4, F8, C3, CP5, T7, T8, O1, O2	**O1**
High	S_3	PO3	AF3, F7, O1, O2, P8	T8, O2, PO3	F3, F7, **O1**
	S_5	O2	Fp1, AF3, F3, FC5, FC1, CP5, CP1, **T7**, O2, P3, P7, PO3	CP1	**O2**
	S_8	**F3**, FC1, C4, T8	C4, T7, O2	**C3**, CP5, **CP1**, T7, P7, **PO3, P4, P8**	O2, P7, PO3
	S_23	Fp1, **F3**, F7, FC5, CP1, CP6, **T7**, O1, PO3	**C3**, CP5, **T7**, O1, O2	AF3, F3, FC5, P8	T7, **T8**
	S_30		T7, O2		**O1**

the changes of the brain activity in the electrode e during emotions at the lower level of the dimension d. For every cluster, we built a circle that encompasses most of the elements of the cluster. It is not necessary that the circle includes all elements of the cluster. The percentage of elements of a given cluster that should be embedded in the corresponding circle should exceed a given threshold. If the two circles intersect this implies that the clustering process hardly separates the two sets of rhythmic brain changes; then, the electrode e is not considered as relevant and it is excluded. We present a sample of tuning in Fig.3. In the sample, subject_1 has three significant electrodes (AF4, F4, and T8) in the alpha band for the lower level of valence and one significant electrode (O1) in the alpha band for the higher level. Using a tuning threshold $= 85\%$, we excluded AF4 and T8 because the two circles of the corresponding clusters intersected. Fig.3 shows the clusters of all relevant electrodes of subject_1 for the valence dimension in the alpha band.

F. CLASSIFICATION

For a novel trial t of a given subject s, we aim to infer its corresponding emotional state η by the following processes: (1) determining the level to which the trial t belongs in every dimension of the VAD model; (2) mapping these levels into an emotional state. As depicted in Fig.4, the classification problem, on which we have focused in this paper, is complex and is addressed in three phases. The first phase aims to predict if a given trial belongs to a given level of a given dimension or not. For every dimension of the VAD model, two one-class classifiers are defined. The first one predicts if the trial belongs to the lower level of the dimension, while the second one predicts if the trial belongs to the upper level of the dimension. These one-class classifiers are trained using a set of trials of the corresponding level and certain trails of the neutral and opposite levels.

The second phase deals with a multi-class classification problem. It aims to predict the level to which the given trial belongs to in every dimension of the VAD model. The two different one-class classifiers of a given dimension vote to reach the objective of this phase. Table 1 presents how the decision is taken based on the classification of the two one-class classifiers of the same dimension.

The third phase aims to predict the emotional state of the trial t from an ensemble of possible emotional states. The final label is decided using the results obtained in the second phase. Table 2 shows the mapping of dimension levels to emotions. Each dimension classifier produces a label, and we decided on the emotion depending on the three labels.

VI. EVALUATION

The proposed method was evaluated using the DEAP dataset. Nine emotions were classified using the VAD model. Previous studies have used a fixed number of electrodes; however, in our approach the choice of electrodes has changed from subject to subject, from one dimension to another, and from

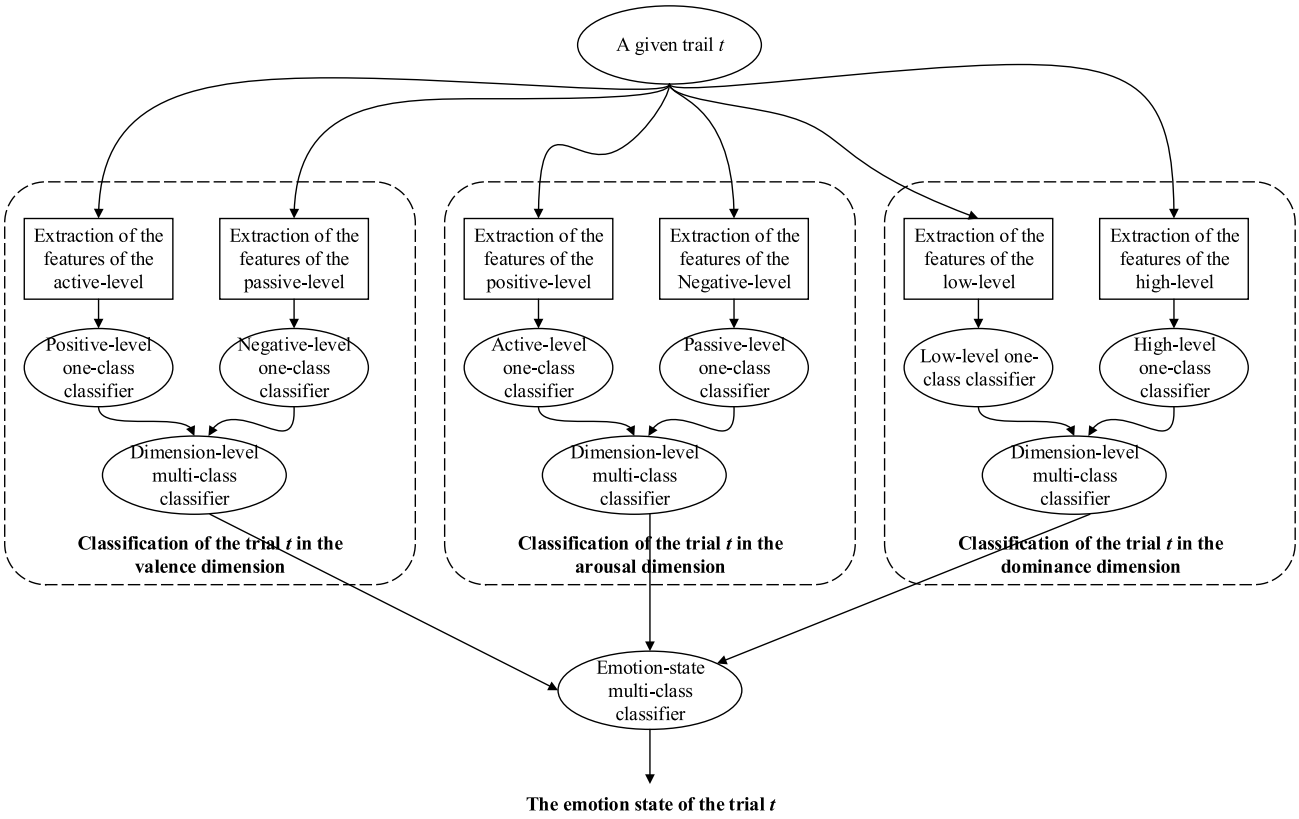


FIGURE 4. The classification process.

TABLE 6. Thresholds setting.

Threshold	Value
$\alpha_threshold$	0.7
$\alpha'_threshold$	0.7
$tuning_threshold$	0.85

one level to another. Our results were compared with those of previous related studies. To test the performance of the proposed work for emotion detection, various metrics were used. These metrics include accuracy, sensitivity (recall), precision, and F1 measure. Statistical tests were also conducted in our study, such as mean, standard deviation, and correlation measurement. All these metrics were used to measure the performance of our proposed approach and to benchmark it corresponding to previous studies.

A. OVERVIEW OF THE DATASET FOR EMOTION ANALYSIS USING PHYSIOLOGICAL SIGNALS

The DEAP dataset is one of the famous datasets in this field [25]. It contains records of the state of a person while watching a musical video. The experiment for DEAP was conducted on 32 subjects. Each subject has recordings of 40 videos (40 trials). They used 40 channels (32 were EEG electrodes and the rest were peripheral channels) that recorded at a sampling rate of 512 Hz, which was then down sampled to 128 Hz. The subjects decided about valence, arousal, dominance, and likes/dislikes using a self-assessment questionnaire. Each subject viewed an array of videos/trials with four labels (arousal, valence, dominance, and liking/disliking) [25]. We mapped the dimensions to distinct emotions. Therefore, there are nine emotions related to the DEAP recordings: happy, pleasant, relaxed, calm, excited, neutral, distressed, miserable, and depressed [25].

B. DATA AUGMENTATION

As mention above, the number of trials per subject related to a given level of a given dimension was very low. To increase the number of trials, we split every trial into fragments of equal window size. Let us consider a trial t of a total time T. Let us consider that W is the window size (the total time) of a single fragment. Let us also consider that the overlapping ratio is O. This ratio determines how much data are common to two consecutive fragments. Every trial t will be split into N fragments of window size W. N is calculated as follows (eq 12):

$$\begin{aligned} N &= \left(\frac{(T-W)}{shift}\right) + 1 \\ shift &= (1-O)W \end{aligned} \tag{12}$$

TABLE 7. Detailed results of the proposed method.

Dimension	Level	Accuracy (%)	Precision (%)	Recall (%)	TPR (%)	TNR (%)	F1- score (%)	Final decision accuracy (%)
Valence	Positive	86.03	92.72	80.72	91.34	80.72	83.88	82.35
	Neutral	84.77	79.76	91.45	84.24	78.52	85.21	
	Negative	85.26	92.67	75.88	94.65	75.88	82.39	
Arousal	Active	78.40	90.13	64.38	92.42	64.38	71.45	79.95
	Neutral	82.08	76.24	90.81	83.16	74.05	82.89	
	Passive	84.27	91.97	76.78	91.76	76.78	82.05	
Dominance	Low	80.85	90.75	72.24	89.47	72.24	76.96	71.14
	Neutral	83.41	79.44	98.75	79.95	74.47	83.89	
	High	82.79	89.43	77.06	88.53	77.06	80.05	

TABLE 8. Comparison between the proposed method and related works.

Method	Number of classes	Feature extraction algorithm	Number of electrodes	Valence (%)	Arousal (%)	Dominance (%)
[16]	2 dimensions with 2 labels for each: Valence (high, low) Arousal (high, low)	Maximum Relevance Minimum Redundancy (MRMR) then principal component analysis (PCA)	32	77.2	74.3	–
[17]	2 dimensions with 2 labels for each: Valence (high, low) Arousal (high, low)	MRMR	32	69.97	71.23	–
[24]	2 dimensions with 2 labels for each: Valence (high, low) Arousal (high, low)	Empirical mode decomposition (EMD) and Intrinsic Mode Functions (IMFs)	(fixed) 8	69.10	71.99	–
[18]	2 dimensions with 2 labels for each: Valence (high, low) Arousal (high, low)	combination of Welch's t-test with the PCA	(fixed) 12	69.6	67.7	–
[19]	3 dimensions with 2 labels for each: Valence (low, high) Arousal (low, high) Dominance (low, high)	Time, frequency and wavelet features	32	56.25	62.34	64.22
[8]	3 dimensions with 3 labels for each: Valence (low, mid, high) Arousal (low, mid, high) Dominance (low, mid, high)	Discrete wavelet transform	32	67.5	69.6	65.1
[21]	3 dimensions with 2 labels for each: Valence (low, high) Arousal (low, high) Dominance (low, high)	Daubechies wavelet transform coefficients	32	73.43	72.65	69.3
[20]	Three different levels of valence and arousal	PSD - Covariate shift adaptation of PCA	32	49.52	46.03	–
Proposed method	3 dimensions with 3 labels for each	The proposed approach	adaptive	82.35	79.95	71.14

In our experiment, we set the window size of the fragments W to 12 s and the overlapping ratio O to 50%. Further, each trial t was split into eight fragments. Therefore, the number of trials was augmented eight times. We considered that the overlapping ratio that was set in this study remains reasonable as 50% of every two consecutive trials were distinct.

Every trial is mapped from a time-domain to a frequency-domain representation. As such 2 trials with different lengths (different epochs) will have 2 frequency-domain representations but with the same number of spectrum. Hence they are processed uniformly regardless of their lengths. The risk of reducing the length of trials is that some (few) trials do not

include significant information. However, the major benefit of decomposing a trial to small epochs is to increase the number of samples that can be more representative of the emotional state [26].

C. RESULTS AND DISCUSSION

Table 3 lists, for certain subjects who were chosen arbitrarily, the electrodes that are relevant to the valence dimension with respect to the frequency bands. The electrodes marked in bold are eliminated by the tuning process.

Table 4 lists, for the same subjects, the electrodes that are relevant to the arousal dimension with respect to the frequency bands. The electrodes that are eliminated by the tuning process are marked in bold.

Table 5 lists after applying the tuning process, for the same subjects, the electrodes that are relevant to the arousal dimension with respect to the frequency bands. The electrodes that are eliminated by the tuning process are marked in bold.

The results of electrodes selection and tuning comply with neuroscience studies [3], [12], [13] that are mentioning that the folding of the cortex differs between any two persons, even monozygotic twins and that the activity is sophisticated and unique in the brain of each person. Indeed, every subject has a distinct and unique brain activity that leads to choosing different electrodes.

The thresholds in this work are listed in Table 6.

The $\beta_threshold$ is computed in an adaptive way. For a given subject s, a given level l of a given dimension d with regards to a given frequency band f, $\beta_threshold$ is calculated according to the following expression (eq 13):

$$\beta_threshold = \frac{\sum_{b_i \in B} \frac{\sum_{t_j \in \varphi(s,d,l)} Avg_\Delta(b_i, t_j, \mathcal{M}_0^{s,d}, f)}{\|\varphi(s,d,l)\|}}{\| B \|}$$
$$\text{where } \| B \| \text{ is the cardinality of set } B \quad (13)$$

Table 7 shows various metrics including the accuracy, recall, precision, F1-score, true positive rate (TPR) and the true negative rate (TNR) that were used to measure the performance of the proposed method. The average accuracy of the proposed method attained 82.35%, 79.95% and 71.14% for the dimensions valence, arousal and dominance respectively. These results were obtained using 10-fold cross validation.

The proposed method is equivalent to selecting the best frequency bands and electrodes for each subject. For benchmarking purpose, we compared our results with previous EEG-based studies, that were tested using the same dataset, including those adopting similar techniques such as PCA and MRMR [16]–[18], [20] for feature extraction. Table 8 shows, for every related work, the number of classes considered in the study (dimensions), the adopted feature extraction algorithm, the number of electrodes used to extract the features, and the average accuracy for the 32 subjects for every dimension.

Compared to the best performance of previous works, the proposed method enhances the system accuracy of both valence and arousal dimensions by 5%. Besides, it enhances the system accuracy of the dominance dimension by 2%.

The average accuracy obtained by the proposed method, to classify the nine emotions of the DEAP dataset, for the overall 32 subjects reaches 65%. We notice that the obtained accuracy exceeds 70% for 40% of the subjects. Furthermore, it exceeds 80% for 20% of the 32 subjects.

VII. CONCLUSION

The VAD constructionist model has emerged as the most reliable model to study emotions as it reduces considerably the complexity of classifying a wide range of emotions. As such, the interest of researchers to the VAD model has significantly increased in the last decade. In this paper we proposed a novel method to classify emotions based on the dimensions of the VAD model. First we explored the cerebral activity by analyzing EEG signals to identify for every subject, in every frequency band, the brain lobes that are sensitive to emotions of the upper- and lower-level of every dimension. Next, the electrodes of the identified brain lobes were tuned to exclude those that had similar firing in both upper- and lower-levels of the corresponding dimension. Changes of brain activity recorded by these electrodes were selected as features.

We proposed a three phases classification method. In the first phase a set of one-class classifiers were defined. For every dimension of the VAD model, two one-class classifiers were defined. The first one predicts if a trial belongs to the lower level of the dimension, while the second one predicts if the trial belongs to the upper level. In the second phase, the results of two one-class classifiers of the same dimension were utilized to predict the level of the dimension which the trial corresponds to. Finally, the results of the second phase were mapped in the third phase into emotions.

The proposed method was tested using the DEAP dataset. Compared to the best performance of previous works, the proposed method enhanced the system accuracy of both valence and arousal dimensions by 5%. Besides, the accuracy of the dominance dimension was improved by 2%.

EEG based emotion recognition remains a complex task and requires multidisciplinary skills including neuroscience, signal processing, machine learning and psychology compared to the image processing based approach. However, conventional algorithms for emotion detection using EEG signals require further exploration in neuroscience and psychology or adopting a multi-modal approach that combines EEG-based emotion recognition models with image processing based approaches to go beyond their current performance.

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